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MEMS Vibration Energy Harvesting for Sustainable Sensor Power: Transduction, System Integration, and Life-Cycle Trade-Offs

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Abstract: Distributed sensing in energy systems requires reliable power supplies where wiring and battery replacement are difficult. Micro-electro-mechanical systems (MEMS) vibration energy harvesters can supply local power, but their sustainability depends on usable energy delivery, operational reliability, and life-cycle burdens. This review examines the conditions under which MEMS vibration energy harvesting can support sustainable sensor operation. Foundational studies and journal literature available by 30 June 2026 were critically examined across piezoelectric, electrostatic, electromagnetic, and triboelectric conversion. Device evidence was assessed alongside fabrication, packaging, excitation conditions, power management, storage, and application validation. Reported outputs were distinguished by measurement boundary, and environmental assessments were examined for their functional units, assumptions, and exclusions. The comparison identified net energy delivery over complete sensing cycles as the basis for assessing autonomous operation. Low-frequency and broadband strategies extended operating ranges under specific excitation conditions, while interface losses, cold-start requirements, and intermittent inputs constrained usable energy. Reliability evidence remained dependent on test conditions, and environmental findings from materials or larger harvesters provided limited grounds for conclusions about complete MEMS sensor nodes. These findings indicate that sustainable sensor power requires coordinated design of the harvester, interface, storage, and sensing workload. The review establishes comparison criteria linking device performance to delivered energy, service continuity, and life-cycle assessment, and identifies the measurements needed to substantiate claims of battery replacement and environmental improvement.

Keywords: MEMS vibration energy harvesting; Autonomous sensors; Net energy delivery; Power management; Reliability; Life-cycle assessment

1 Introduction

Distributed sensors support condition monitoring in energy equipment, industrial machinery, and infrastructure, but their power supplies can limit the duration and practicality of deployment. Wiring may be difficult at inaccessible or moving locations, while battery replacement introduces maintenance requirements that depend on installation conditions and sensing workloads. Vibration energy harvesting offers a local power source where mechanical motion accompanies the process being monitored. Its relevance to energy sustainability therefore rests on whether the harvested energy can sustain a useful sensing service and reduce the resources required to provide that service [1].

Micro-electro-mechanical systems (MEMS) provide a fabrication platform for compact vibration harvesters and their integration with sensing components. Piezoelectric, electrostatic, electromagnetic, and triboelectric mechanisms offer different routes from mechanical motion to electrical output, each with distinct material, structural, and interface requirements. Miniaturisation, however, constrains proof mass and displacement, while the available vibration varies with the operating state and mounting location of the monitored equipment. A successful device must consequently match both its mechanical environment and the electrical demands of the node. High output at a selected laboratory resonance is only one part of that requirement.

Recent reviews provide substantial foundations for examining these choices. He and Briscoe [2] reviewed piezoelectric materials, conversion mechanisms, and multifunctional applications, while Sabzpoushan and Woias [3] examined electret-based harvesters. An et al. [4] considered structural superlubricity and triboelectric generation in MEMS, including the challenges associated with their integration. Complementing these mechanism-focused accounts, Zhuo et al. [5] reviewed frequency up-conversion components, and Lu et al. [6] assessed fatigue in vibration energy harvesters. Table 1 summarises these complementary perspectives. Together, they establish the need to connect material and device advances with the operating conditions under which those advances deliver useful energy.

Table 1. Selected recent reviews and their relevance to the present review

Reference	Principal Focus	Connection to the Present Review
Li et al. [1]	Vibration harvesting for self-sustained sensing systems	Application and system context
He and Briscoe [2]	Piezoelectric materials, mechanisms, and applications	Material–transducer relationships
Sabzpoushan and Woias [3]	Electret-based energy harvesters	Electrostatic conversion and electret selection
An et al. [4]	Superlubricity and triboelectric generation in MEMS	Integration and contact-related constraints
Zhuo et al. [5]	Frequency up-conversion components	Coupling low-frequency inputs to resonant structures
Lu et al. [6]	Fatigue in vibration energy harvesters	Durability and failure mechanisms

Note: MEMS = micro-electro-mechanical system.

This connection requires consistent treatment of the complete energy path. Power measured across a matched resistor does not establish the net energy delivered to a sensing load over a complete operating cycle. The sensing workload also matters: periodic measurements and transmissions impose different requirements from continuous acquisition, particularly when vibration is intermittent. Bouhedma et al. [7], for example, investigated battery-assisted and battery-free operation of an environmental sensor node under controlled harmonic excitation. Such demonstrations help establish feasible operating scenarios, but short laboratory tests cannot establish long-term autonomy under changing field conditions. Device performance and deployment readiness must therefore be assessed separately.

Environmental performance introduces a further boundary. Replacing some battery capacity does not by itself establish a net environmental improvement, because the harvester, package, interface electronics, and storage components also require materials and manufacturing. The carbon-footprint assessment of lead zirconate titanate (PZT) components and road-integrated piezoelectric systems by Sharafi et al. [8] illustrates the importance of specifying the assessed system. At the material level, Franco et al. [9] compared lead-free piezoelectric composites with PZT under defined production and use assumptions. These studies inform assessment methods, but their results cannot be transferred directly to complete MEMS sensor nodes. Environmental conclusions must retain the original functional units, component boundaries, and lifetime assumptions.

Against this background, this review examines MEMS vibration harvesting through three linked questions: how much net energy reaches the sensing load, what evidence supports reliable operation over the intended service life, and under which conditions environmental gains are substantiated. Foundational studies and journal literature available by 30 June 2026 are considered within this scope. The analysis preserves the distinction between MEMS devices and larger systems used to demonstrate transferable concepts, and between measured performance, model predictions, and prospective applications.

Section 2 defines the review scope and evidence rules, and Section 3 establishes vibration-resource and sensor-energy requirements. Sections 4–6 examine transduction, fabrication and reliability, and operation under variable excitation. Sections 7 and 8 connect power management to application evidence, before Section 9 evaluates life-cycle trade-offs and research priorities. Section 10 draws together the conditions required for sustainable sensor power.

2 Review Scope, Search Strategy, and Evidence Assessment

2.1 Scope and Eligibility

This review adopts a critical narrative approach to examine the relationship between MEMS vibration harvesting and sustainable sensor power. Its scope follows the three questions established in Section 1: energy delivery to the sensing load, reliability over the intended service life, and the conditions supporting environmental improvement.

Piezoelectric, electrostatic, electromagnetic, and triboelectric conversion form the device-level foundation, with power management, storage, packaging, and sensing workloads included where they affect these questions.

Microfabricated harvesters and devices explicitly intended for MEMS integration received priority. Larger harvesters were retained when they provided relevant evidence on excitation strategies, interface operation, application validation, or environmental assessment. Their scale and input source were distinguished from those of MEMS devices. Studies using other energy sources were considered only where their circuit or energy-management principles informed the analysis; their outputs were not treated as vibration-harvesting performance.

2.2 Targeted Literature Update

The original manuscript provided the starting bibliography. Its references were reassessed for relevance, and a targeted update prioritised journal literature from 2023 onwards while retaining foundational studies needed to explain established models and mechanisms. Searches combined "MEMS" or "microfabricated" with "vibration energy harvesting" and mechanism-specific terms, followed by terms addressing frequency up-conversion, interface circuits, cold start, storage, fatigue, packaging, autonomous sensing, and life-cycle assessment. Publisher pages, DOI records, accessible full texts, and author or institutional repositories were consulted to verify bibliographic information and relevant claims.

The literature cutoff was 30 June 2026. Articles assigned to later issues were eligible only when a formal online publication date on or before that cutoff could be verified. The update was checked in October 2026; this verification date was kept separate from the publication cutoff. Journal-level Web of Science or Scopus coverage was checked using publisher, society, or institutional records. These checks did not constitute article-level database accession verification. As no reproducible database-export and screening log was available, the review does not claim an exhaustive systematic search.

2.3 Evidence and Comparability Rules

Evidence was evaluated at the level of the claim it could support. Abstracts and publication records supported bibliographic verification and limited qualitative descriptions. Numerical comparisons required examination of the relevant original methods and results. Where a necessary condition remained unresolved, the numerical claim was omitted or explicitly qualified; missing information was not inferred.

Device comparisons distinguished excitation frequency from structural resonance and recorded acceleration conventions, load conditions, and output boundaries where verified. Bare-device power, conditioned output, stored energy, and energy delivered to a functioning node were treated separately. Incomplete test conditions precluded normalisation or cross-device ranking. Likewise, simulations, laboratory demonstrations, field measurements, and projected lifetimes were identified as different forms of evidence.

Environmental findings were assessed with reference to functional units, component boundaries, manufacturing assumptions, service life, and end-of-life treatment. Material-level results were not extrapolated to complete sensor nodes. These rules provided the basis for the following sections: first defining the available vibration and sensing demand, then assessing whether device and system evidence supported the proposed operating conditions.

3 Vibration Resources and Energy Requirements of Autonomous Sensors

3.1 Source Spectra and Displacement Constraints

The suitability of vibration harvesting depends on the motion available at the installation point. Early measurements by Roundy et al. [10] established low-level environmental vibration as a potential power source for wireless sensors, while Mitcheson et al. [11] examined the constraints associated with harvesting human and machine motion. These studies remain useful foundations, but their source measurements cannot substitute for characterisation of the intended deployment. Equipment speed, loading, mounting stiffness, and operating schedules can change both the vibration spectrum and the duration for which energy is available.

Frequency and acceleration must therefore be considered together. For sinusoidal base motion, the displacement amplitude is:

$$Y = \frac{A}{(2\pi f)^2} \tag{1}$$

where, Y is the base-displacement amplitude, A is the corresponding acceleration amplitude, and f is the excitation frequency. The displacement and acceleration must use the same amplitude convention. Eq. (1) describes the base motion; it does not determine the relative displacement of the harvester proof mass, which also depends on resonance, damping, and mechanical limits. For broadband or transient inputs, a single acceleration–frequency pair is insufficient, and time histories or spectral descriptions are needed.

This distinction matters particularly at the MEMS scale, where proof mass and available travel are constrained. A resonant structure can amplify relative motion near its natural frequency, but frequency drift can reduce that

response, while displacement limits can restrict operation under stronger excitation. Transformer temperature monitoring provides a directly relevant energy-sector example: Wang et al. [12] investigated a piezoelectric harvesting system under low-acceleration conditions. The reported operating scenario demonstrates feasibility for transformer monitoring under the tested excitation and sensing schedule. Similarly, Peralta-Braz et al. [13] used measured bridge vibrations in a numerical framework for joint optimisation of energy harvesting and sensing. Their numerical framework links measured bridge vibrations to joint optimisation of harvesting and sensing.

3.2 Sensor Duty Cycles and Energy Budgets

Once the input is characterised, the electrical demand must be defined at the level of the sensing service. A node may spend most of its time asleep yet require substantially more power during acquisition, processing, or radio transmission. Average demand consequently depends on task frequency, execution time, and communication behaviour. Batteryless computing makes this dependence explicit because intermittent energy availability can determine when a task starts and whether it completes [14].

A useful accounting boundary is the storage element. Its energy balance can be written as:

$$\frac{\mathrm{d}E_{\mathrm{s}}}{\mathrm{d}t} = P_{\mathrm{ch}}(t) - P_{\mathrm{draw}}(t) - P_{\mathrm{leak}}(t) \tag{2}$$

where, E_{s} is stored energy, P_{ch} is power entering storage after upstream conversion losses, P_{draw} is power withdrawn to operate the node, including downstream conversion and control consumption, and P_{leak} represents storage leakage. These definitions prevent bare-transducer output from being treated as usable load power and avoid counting the same conversion loss twice.

Over a representative interval, a non-negative energy balance is necessary for operation without progressive depletion. It is not sufficient, however, because an energy deficit can occur before a later harvesting period restores the average balance. The usable storage capacity must bridge such intervals, and its voltage and current capability must support individual tasks. Chamanian et al. [15] demonstrated energy-neutral operation through adaptive duty-cycle control in a vibration-powered wireless sensor network. This establishes a practical means of matching demand to supply, while also showing why the achieved sensing and transmission rates must accompany claims of autonomy.

3.3 Design Requirements

The input and load descriptions together define the design problem. Mechanical characterisation should establish the available spectrum, amplitude convention, direction, variability, and inactive periods. Electrical characterisation should specify task energy, standby demand, startup requirements, and the minimum acceptable sensing service. Neither description is complete without the other: a higher harvesting rate has limited value if startup remains unreliable, while a lower duty cycle may balance energy at the expense of the monitoring requirement.

Energy-management studies provide useful guidance on linking the source, storage, protection, and load. Monagle et al. [16], for example, presented such a design framework using magnetic-field harvesting. Its energy-management principles inform the present discussion, although its prototype output is not evidence of MEMS vibration-harvester performance.

Figure 1 places these requirements within the complete energy path, from mechanical excitation to the sensing workload. The storage element buffers differences between generation and demand, while interface and storage losses determine how much energy remains available to the node. The control link represents adjustment of the electrical operating point and workload rather than an additional energy source.

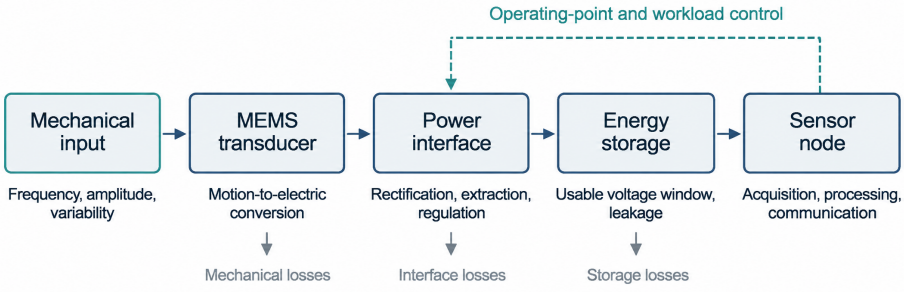


Figure 1. Energy flow and control in a vibration-powered sensor node

Note: MEMS = micro-electro-mechanical system.

The resulting assessment must therefore ask whether a device can sustain a specified workload under a defined input and interruption pattern. This criterion provides the basis for Section 4, where the four transduction mechanisms are examined within explicit mechanical and electrical boundaries.

4 Transduction Mechanisms and Comparable Device Performance

4.1 Common Modelling and Comparison Boundary

The requirements established in Section 3 provide the basis for comparing transduction mechanisms. Each harvester must couple an available mechanical input to an electrical load within limits imposed by displacement, fabrication, and integration. The inertial-generator model introduced by Williams and Yates [17] provides a starting point for this analysis. For a single mechanical mode under base excitation,

$$m\ddot{z} + c_{\mathrm{m}}\dot{z} + kz + F_{\mathrm{tr}} = -mj \tag{3}$$

where, m is the effective modal mass, z is displacement relative to the base, y is base displacement, c_{m} is mechanical damping, k is modal stiffness, and F_{tr} is the signed transduction reaction term. Dots denote time derivatives. Eq. (3) assumes linear mechanical stiffness and damping; nonlinear restoring forces, impacts, and displacement stops require additional terms or a different formulation. The electrical response is coupled to the mechanical motion through F_{tr}, so the load can alter displacement as well as extracted power.

Architectural analysis by Mitcheson et al. [18] further established the importance of displacement constraints when selecting vibration-driven generators. Accordingly, neither resonant frequency nor acceleration alone determines the preferred mechanism. Comparisons must retain the excitation waveform, amplitude convention, available travel, electrical load, and measurement boundary. Device dimensions must also identify whether they describe the active structure, chip, or assembled package.

These distinctions are retained below. Historical devices establish the development of each mechanism, while recent studies illustrate specific advances. Their outputs are not ranked across unmatched test conditions.

Figure 2 shows representative structural arrangements for the four transduction mechanisms. The sketches identify the motion and conversion elements used in the following discussion; they do not specify a common device size or a fabrication-equivalent architecture.

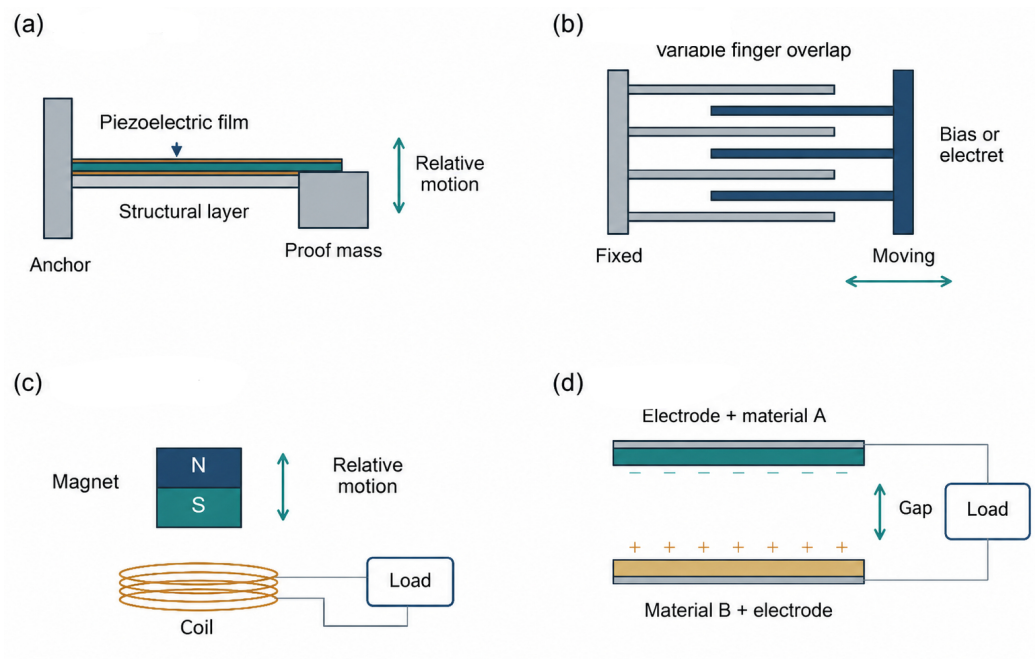


Figure 2. Conceptual structures of (a) piezoelectric, (b) electrostatic, (c) electromagnetic, and (d) triboelectric harvesters (not to scale)

4.2 Piezoelectric Conversion

Piezoelectric harvesters convert deformation of a polarised material into electrical charge. In bending structures, the position of the active film relative to the neutral axis and the distribution of strain determine how effectively

mechanical motion contributes to charge generation. Electrode coverage and the electrical connection of multiple beams also influence the output. Material coefficients therefore describe only part of the conversion process.

Erturk and Inman [19] developed a distributed-parameter model for piezoelectric cantilevers that accounts for their structural and electrical behaviour. For an isolated mode, a reduced representation is:

$$\begin{aligned} m\ddot{z} + c_{\mathrm{m}}\dot{z} + kz + \theta v &= -m\ddot{y}, \\ C_{\mathrm{p}}\dot{v} + \frac{v}{R} - \theta\dot{z} &= 0 \end{aligned} \tag{4}$$

where, v is terminal voltage, C_{p} is the effective piezoelectric capacitance, R is a resistive load, and θ is the modal electromechanical coupling coefficient. The equations use consistent mechanical and electrical polarity conventions. They neglect dielectric leakage and assume linear piezoelectric behaviour with a purely resistive external circuit. Rectification and switching interfaces require a corresponding circuit model.

Early thin-film PZT [20] and aluminium nitride (AlN) devices [21] established important MEMS implementations. Recent work has extended this foundation through alternative films, modified structures, and array connections. A $BiFeO_3$-film harvester combined with an external dynamic magnifier investigated the response to impulsive forces [22]. Its relevance lies in coupling a MEMS element to an auxiliary mechanical structure; the resulting assembly must be distinguished from the chip alone.

A scandium-alloyed AlN (ScAlN) cantilever array reported 2.53 µW across a 1 MΩ load at approximately 465 Hz and a stated acceleration of 1 g [23]. The array connection used ten of the twenty fabricated beams to account for their phase relationships. A separate conditioning experiment produced a 1.8 V output at a root-mean-square (RMS) acceleration of 2 g, with no functional load connected. These results demonstrate different stages of the energy path and should not be combined into a claim of powered-node operation.

Material and integration advances address additional constraints. Mn-doped bismuth ferrite films were investigated for improved electromechanical coupling in MEMS harvesters [24], while an AlN device incorporated wafer-integrated magnetic material for mechanical and contactless magnetic excitation [25]. The latter reported 24.75 µW under mechanical excitation at 1230 Hz and 0.75 g. Its magnetic material formed part of the excitation and mass architecture; electrical conversion remained piezoelectric.

4.3 Electrostatic Conversion

Electrostatic harvesters use motion to vary capacitance. Their operation depends on an electrical bias supplied externally or maintained by an electret, together with a circuit that transfers charge during the mechanical cycle. Early work on vibration-to-electric conversion established this variable-capacitance approach [26].

For an ideal capacitor,

$$U_{\mathrm{e}} = \frac{Q^2}{2C} = \frac{1}{2}CV^2 \tag{5}$$

where, U_{e} is stored electrostatic energy, Q is charge, C is capacitance, and V is voltage. Although both expressions describe the same capacitor state, energy changes during motion depend on whether charge, voltage, or another circuit-dependent condition is maintained. Eq. (5) alone therefore does not specify harvested energy per cycle. That quantity requires the complete charging and extraction sequence, including the contribution of any bias source.

Recent research has addressed electret maintenance, nonlinear response, and fabrication. A self-rechargeable electret architecture investigated recovery of the charge supporting conversion [27]. This mechanism is relevant to charge stability, but recovery behaviour does not establish stability over the intended deployment life. A two-sided nonlinear MEMS architecture examined a different route through its dynamic response [28]. Its operating branches and excitation thresholds must remain associated with the model and test conditions under which they were obtained.

Fabrication studies have combined anodic bonding with electret charging [29] and introduced a post-processed self-assembled electret [30]. Such approaches address integration requirements, but eliminating a conventional charging step does not establish negligible manufacturing energy or permanent charge retention. For sensor power, the assessment must also include parasitic capacitance, leakage, bias requirements, and extraction losses. These determine how much of the mechanically induced electrical variation becomes usable energy.

4.4 Electromagnetic Conversion

Electromagnetic harvesters generate an electromotive force through changing magnetic flux linkage. In a simplified linear representation,

$$e = -\frac{\mathrm{d}\lambda}{\mathrm{d}t} = -\Gamma\dot{z} \tag{6}$$

where, e is induced electromotive force, λ is coil flux linkage, and $\Gamma = \mathrm{d}\lambda/\mathrm{dz}$ is the flux-linkage gradient. The final expression assumes that Γ is approximately constant over the relevant travel. Terminal voltage and load power additionally depend on winding resistance, inductance, and the connected circuit.

Microelectromagnetic generators established this mechanism at small scales [31], but miniaturisation constrains both the magnetic circuit and the winding. Increasing the number of turns can increase induced voltage while also changing resistance and fabrication requirements. The resulting design must be assessed as an assembled magnetic and electrical system.

Lecerf et al. [32] integrated neodymium–iron–boron (NdFeB) micromagnets into a MEMS resonator and reported load power of approximately 6 nW near 1225 Hz. The reported acceleration of 0.8 g was estimated from resonator displacement and quality factor. The work provides evidence of micromagnet integration rather than autonomous-node power delivery. The device combines monolithically integrated micromagnets with a separately fabricated coil.

Other studies have investigated permanent–soft-magnetic bistable coupling [33], bistable structures using MEMS three-dimensional coils [34], and printed microstructures assembled with commercial coils and magnets [35]. These approaches address different aspects of response and manufacture. Bistability must be evaluated against the excitation required to access the intended motion, while multifrequency structures require separate accounting of their channels. Outputs measured at different resonances cannot automatically be summed as simultaneous power at one operating point.

4.5 Triboelectric Conversion

Triboelectric devices combine contact electrification with electrostatic induction. Relative motion changes the electrical configuration associated with surface charge, allowing charge transfer through an external circuit. Their response depends on contact conditions, surface properties, separation, and circuit loading.

A triboelectric MEMS accelerometer demonstrated that this mechanism can be implemented for microscale inertial sensing [36]. Nevertheless, a self-generated sensing signal and a supply capable of operating processing and radio electronics are different outcomes. Open-circuit voltage cannot establish the latter, particularly for a source whose terminal response is strongly load dependent.

The performance framework developed for triboelectric nanogenerators (TENGs) [37] supports evaluation of charge transfer and energy over a defined cycle. For a variable-capacitance representation,

$$V(x, Q) = V_{\mathrm{oc}}(x) - \frac{Q}{C(x)} \tag{7}$$

where, x describes the mechanical configuration, Q is transferred charge under the selected sign convention, $C(x)$ is configuration-dependent capacitance, and $V_{\mathrm{oc}}(x)$ is the corresponding open-circuit voltage. The relation assumes a specified surface-charge state. Changes in charge caused by contact history or environmental exposure require further treatment. Average power must be obtained from the loaded voltage and current over a defined interval, rather than by multiplying separately measured open-circuit voltage and short-circuit current.

Origami structures [38] and stretchable silicone-rubber architectures [39] have been investigated for vibration-driven triboelectric conversion. They provide structural concepts relevant to variable inputs, but their larger assemblies do not establish MEMS fabrication or packaged microsystem performance. Translation to MEMS requires assessment of contact mechanics, tolerances, packaging, and wear alongside electrical output.

4.6 Mechanism Selection

The preceding evidence supports selection against a defined operating requirement rather than a universal hierarchy of mechanisms. Piezoelectric conversion couples strain to charge, making film placement and structural deformation central to performance. Electrostatic conversion links capacitance variation to bias retention and charge-transfer control. Electromagnetic conversion depends on the magnetic circuit, relative velocity, and winding losses. Triboelectric conversion introduces contact and surface-charge behaviour into both the electrical response and reliability assessment.

Hybridisation can combine complementary responses, but it also changes the mechanical assembly and electrical interface. A dual piezoelectric–electromagnetic harvester used frequency up-conversion to capture low-frequency human motion [40]. The larger prototype demonstrates an up-conversion strategy whose MEMS implementation requires control of travel, coupling, and packaging. Additional transducers should therefore be justified by their contribution to usable energy under the intended input, together with their integration cost.

Table 2 summarises representative evidence without imposing a cross-device power ranking. Figure 3 plots three reported load-power values from this evidence alongside their individual excitation conditions and measurement boundaries. The points retain their source conditions and do not constitute a common-condition performance benchmark.

Table 2. Representative device evidence and conditions governing comparison

Mechanism and Reference	Device or Integration Feature	Verified Excitation or Operating Condition	Evidence and Comparison Boundary
Piezoelectric [22]	$BiFeO_3$ MEMS element with an external dynamic magnifier	Impulsive excitation; auxiliary metal structure	Supports impulse-response analysis; chip and complete assembly must be distinguished
Piezoelectric [23]	ScAlN cantilever array	Approximately 465 Hz; stated 1 g; 1 MΩ load	Reported 2.53 µW at the load; acceleration convention was not resolved in this review, precluding normalisation
Piezoelectric [24]	Mn-doped bismuth ferrite film	Device-specific electromechanical characterisation	Supports coupling analysis; material and device coupling measures must remain distinct
Piezoelectric [25]	AlN MEMS structure with integrated magnetic material	Mechanical resonance at 1230 Hz and stated 0.75 g	Reported 24.75 µW; mechanical and contactless magnetic excitation results must remain separate
Electrostatic [27]	Self-rechargeable electret	Charge-recovery operation	Supports the recovery mechanism; does not establish deployment-length charge stability
Electromagnetic [32]	Integrated NdFeB micromagnets and separately fabricated coil	Approximately 1225 Hz; reported 0.8 g estimated from displacement and quality factor	Reported load power of approximately 6 nW; no inference of conditioned node power
Electromagnetic [34, 35]	MEMS coils or printed microstructures in assembled devices	Architecture-specific bistable or multifrequency operation	Integration levels and channel outputs require separate accounting
Triboelectric [36]	MEMS accelerometer	Inertial sensing demonstration	Self-generated sensing output does not establish power delivery to a complete wireless node

Note: MEMS = micro-electro-mechanical system; ScAlN = scandium aluminum nitride; AlN = aluminum nitride.

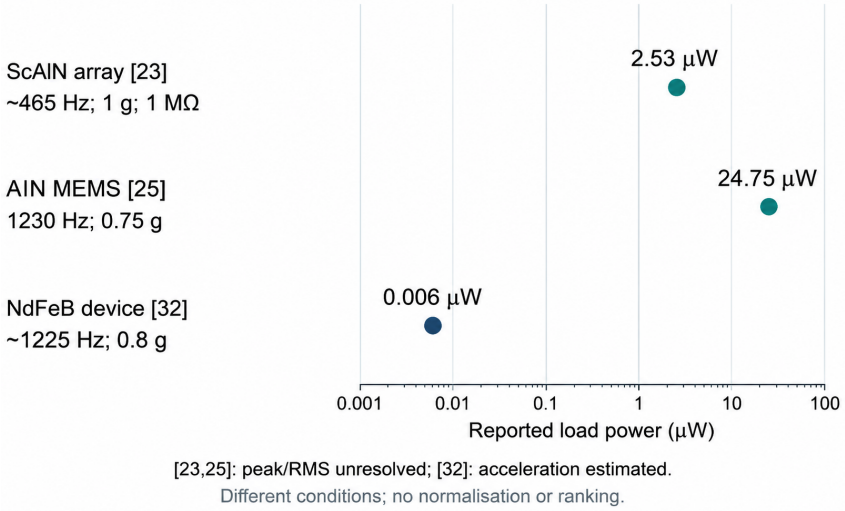


Figure 3. Reported load power under source-specific excitation conditions [23, 25, 32] (values are not normalised or directly comparable)

Note: ScAlN = scandium aluminum nitride; AlN = aluminum nitride; MEMS = micro-electro-mechanical system; RMS = root mean square.

5 Materials, Microfabrication, Packaging, and Reliability

5.1 Thin Films and Material Selection

The mechanism comparison in Section 4 establishes how mechanical motion can generate electrical output. Realising that response in a usable device also requires a material stack that can be fabricated, released, connected, and protected without compromising its function. Material selection must therefore account for residual stress, processing temperature, electrode compatibility, and structural integrity alongside electromechanical properties.

Scandium alloying illustrates the connection between material development and device design. Akiyama et al. [41] established an increased piezoelectric response in ScAlN films, providing a foundation for subsequent devices such as the cantilever array discussed in Section 4 [23]. The material response does not determine array output independently: film stress, beam geometry, electrode layout, and phase relationships remain part of the conversion process. Comparisons based solely on a piezoelectric coefficient consequently omit factors that govern the fabricated structure.

Recent studies have explored different routes to reducing substrate constraints and accommodating flexible structures. Liu et al. [42] developed freestanding lead-free piezoceramic films, while Han et al. [43] deposited amorphous perovskite films directly on plastic substrates without subsequent annealing. These approaches address different manufacturing requirements and should be assessed accordingly. Freestanding films offer opportunities to control mechanical constraint, whereas direct deposition can avoid a separate transfer step. Neither advantage establishes compatibility with every MEMS process or application.

Ren et al. [44] demonstrated flexible harvesters using freestanding ferroelectric oxide films released through a water-soluble sacrificial layer. Their PZT films also show why processing and composition require separate consideration. Water-assisted release does not remove lead from the active material or account for the preceding high-temperature growth steps. Likewise, bending tests on flexible films cannot establish performance under the base-excitation conditions used for MEMS cantilevers. Such studies inform material and fabrication choices, while their environmental implications require the broader assessment developed in Section 9.

5.2 Fabrication and Package Integration

After material selection, the assembly must preserve the mechanical and electrical conditions on which conversion depends. Release processing, proof-mass attachment, electrical interconnection, and mounting can alter the response of a structure characterised earlier in fabrication. Packaging is therefore part of the harvester design. It must provide protection while retaining the clearance and mechanical coupling required for operation.

The relevant integration boundary differs among mechanisms. A deposited piezoelectric film may form part of a microfabricated beam, whereas magnetic or contact-based architectures can require separately produced components. The electromagnetic device discussed in Section 4, for example, integrated micromagnets into its resonator but used a separately fabricated coil [32]. Describing the complete assembly makes its manufacturing contribution clear and prevents chip dimensions from being mistaken for packaged volume.

Triboelectric devices introduce an additional requirement because their electrical response depends on interacting surfaces. Bu et al. [45] investigated a submillimetre-scale superlubric triboelectric generator as a route to reducing friction at the conversion interface. Its significance concerns the behaviour of that interface under specified conditions; the reported characteristic length should not be treated as the dimensions of the complete system. Translation into a packaged sensor supply also requires preservation of surface condition, alignment, and charge behaviour during exposure and repeated motion.

These examples indicate what fabrication comparisons should establish: which components were microfabricated, how they were assembled, and whether performance was measured before or after packaging. This information is needed to judge both reproducibility and the practical consequences of integration.

5.3 Fatigue, Wear, and Charge Stability

Packaging and material choices ultimately affect how performance changes with use. The fatigue review discussed in Section 1 [6] and the experiments of Yang and Ahmad-Mazlan [46] established repeated mechanical loading as a necessary part of harvester assessment. Frequency alone, however, cannot determine lifetime. Strain amplitude, loading history, geometry, and the criterion used to define failure must accompany any comparison.

Interfaces within the material stack also require attention. Ravikumar and Markevicius [47] examined coated poly(vinylidene fluoride) (PVDF) structures under thermal and mechanical loading. Their findings concern the tested material combinations, rather than an isolated coating property that can be transferred directly to silicon MEMS devices. Similarly, the deformation-restricted harvester developed by Lin et al. [48] addressed excessive motion through a stopper-based design. Restricting deformation provides a means of managing mechanical loading, but its value for lifetime must be demonstrated through the resulting operating stresses and endurance behaviour.

For electrostatic harvesters, mechanical survival must be considered alongside charge retention. Sabzpoushan et al. [49] investigated protective layers for spontaneously polarised electrets, addressing the exposure sensitivity of the charge-supporting material. Protection introduces its own design balance because the added layer also affects the electrical configuration. Environmental and repeated-impact tests characterise electret stability under the reported exposure and loading conditions.

Table 3 brings these issues together. Reliability evidence is most useful when it reports changes in output as well as structural survival, with exposure conditions and failure criteria stated explicitly. The same principle applies to low-frequency operation: increasing travel or introducing impacts may improve mechanical coupling while changing the loading history. Section 6 examines those strategies in relation to both energy capture and their operating constraints.

Table 3. Material, fabrication, and reliability evidence relevant to MEMS vibration harvesting

Design Route	Evidence	Implication for Assessment
ScAlN films [41] and MEMS arrays [23]	Material response and device implementation	Relate film properties to structural stress and electrical connection
Freestanding lead-free films [42] and direct deposition on plastic [43]	Alternative film-processing routes	Assess release, transfer, and substrate compatibility separately
Freestanding PZT films [44]	Flexible structures using sacrificial-layer release	Retain composition and upstream processing within the environmental boundary
Superlubric triboelectric interfaces [45]	Friction-related interface design	Preserve test atmosphere, loading, and dimensional definitions
Fatigue and coated-polymer studies [46, 47]	Endurance under specified loading	Compare strain, exposure, material stack, and failure criteria
Deformation restriction [48]	Stopper-based control of motion	Distinguish mechanical protection from demonstrated lifetime
Protected electrets [49]	Charge stability under tested exposure	Evaluate protection together with its electrical consequences

Note: MEMS = micro-electro-mechanical system; ScAlN = scandium aluminum nitride; PZT = lead zirconate titanate.

6 Operation Under Low-Frequency and Variable Vibrations

6.1 Nonlinearity and Tuning

The fabrication and reliability constraints discussed in Section 5 become particularly important when the excitation departs from steady resonance. A structure designed for one frequency may receive little useful excitation during changes in equipment speed or loading. Low-frequency inputs introduce a related difficulty: coupling slow motion to a compact transducer without exceeding its available travel. Frequency up-conversion, established in early microsystem research by Kulah and Najafi [50], addresses this mismatch by separating the input frequency from the subsequent transducer oscillation. Nonlinear structures, multiple modes, and tuning offer other routes, each with different operating requirements.

Figure 4 distinguishes the physical basis of these strategies. Tuning shifts the resonant operating point, multiple modes provide distinct response peaks, and bistability introduces an energy barrier between stable configurations. Frequency up-conversion instead separates the slow triggering input from the subsequent resonator oscillation.

Nonlinear harvesting replaces the linear stiffness term kz in Eq. (3) with a nonlinear restoring-force relation. Cottone et al. [51] demonstrated the potential of nonlinear dynamics for energy harvesting, providing a foundation for subsequent architectures. Their relevance depends on the motion actually reached under the available excitation. In a bistable structure, oscillation within one potential well differs from motion between wells; a response requiring stronger excitation cannot be assumed at a lower-amplitude installation. Initial conditions and sweep direction can also affect measured response. A wider frequency-response curve therefore needs evidence that the intended branch remains accessible during operation.

Multiple-mode architectures address frequency mismatch through a different structural route. Feng et al. [52] investigated a heterogeneous cascaded MEMS harvester with integrated PVDF films for sub-10 Hz excitation. This provides a recent example of low-frequency MEMS development, although it does not establish net power delivery to a sensor node. He et al. [53] examined a multimode auxetic piezoelectric architecture using a larger printed structure. Its contribution concerns structural response rather than demonstrated silicon-MEMS fabrication. In both cases, the

useful operating range depends on how strongly the input excites each mode and how the resulting electrical outputs are collected.

Tuning instead seeks to maintain a favourable operating point as conditions change. Tsuboi et al. [54] developed an automatically tracking electret harvester, while Decroix et al. [55] investigated microcontroller-based maximum-power-point tracking using rapid impedance measurement. These approaches act on different parts of the coupled system: frequency tracking and electrical operating-point adjustment should not be treated as interchangeable. Their value must be assessed after accounting for sensing, control, and adjustment consumption over the relevant operating interval.

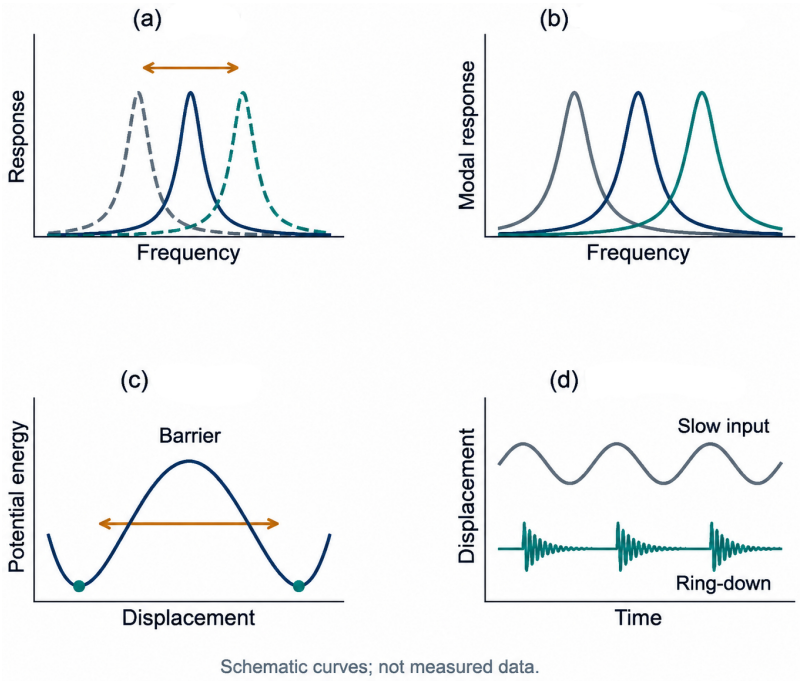


Figure 4. Schematic illustrations of (a) resonance tuning, (b) multiple modes, (c) bistability, and (d) frequency up-conversion

6.2 Frequency Up-Conversion

Frequency up-conversion allows a slow input to trigger faster motion of an energy-converting element. Contact, plucking, or magnetic interaction can transfer energy to a resonator, which then undergoes a decaying oscillation. The review by Zhuo et al. [5] provided a classification of these components. For device comparison, the trigger frequency, resonator frequency, and energy transferred per event must remain distinct.

This separation can improve coupling to slow motion, but it introduces a transfer stage whose losses and loading affect performance. Mechanical contact may change wear and impact stresses, whereas magnetic coupling requires additional components and a suitable interaction geometry. In the integrated piezoelectric device discussed earlier [25], mechanical and contactless magnetic excitation produced different operating conditions. Its low-frequency mechanical tests used high-acceleration pulses, which cannot be equated with weak, continuous base vibration.

The larger piezoelectric–electromagnetic prototype in the work of Shi et al. [40] likewise demonstrated an up-conversion strategy under low-frequency human motion. Translation to MEMS requires reassessment of travel, tolerances, coupling, and packaging. The relevant comparison is whether the additional transfer stage increases energy delivered under the intended input, while preserving the required service life.

6.3 Impulsive and Intermittent Excitation

Impulsive inputs require a time-domain assessment because their energy contribution depends on both the response to each event and the interval between events. The $BiFeO_3$ MEMS harvester with an external dynamic magnifier [22] illustrates an architecture designed to modify the response to short mechanical inputs. Its auxiliary structure belongs within the mechanical and dimensional boundary of the evaluated assembly.

Neither peak voltage nor energy from one impulse establishes average supply power. The event rate, ring-down, electrical extraction, and losses between events determine the longer-term energy balance. Closely spaced events

may interact with residual oscillation, while long inactive periods increase the importance of storage leakage and standby consumption. Repeated impulses also change the loading history relevant to fatigue and contact reliability.

Table 4 summarises the resulting assessment requirements. Across these strategies, bandwidth should be defined by an explicit response or usable-output criterion, rather than by the appearance of additional resonances alone. The decisive question remains whether adaptation increases energy available to the sensing workload under a representative input history. Section 7 examines the electrical interfaces and storage arrangements needed to establish that result.

Table 4. Strategies for low-frequency and variable excitation and their assessment requirements

Strategy	Representative Evidence	Conditions Governing Useful Operation
Nonlinear and bistable response	[28, 33, 34, 51]	Accessible response branches, excitation thresholds, initial conditions, and sweep history
Cascaded and multimode structures	[52, 53]	Modal participation, input coupling, device scale, and collection of separate outputs
Automatic tracking and electrical adaptation	[54, 55]	Tracking range and dynamics, together with measurement and control consumption
Frequency up-conversion	[5, 25, 40, 50]	Trigger and resonator frequencies, transfer losses, component dimensions, and repeated loading
Impulse-response modification	[22]	Pulse duration, event rate, auxiliary structures, extracted energy, and inactive-period losses

7 Power Management, Storage, and Net Energy Delivery

7.1 Rectification and Extraction

The strategies discussed in Section 6 can change the mechanical response, but their value to a sensor depends on the energy delivered through the electrical interface. Recent reviews distinguish rectification, synchronous switching, charge extraction, and operating-point control as related but separate functions [56, 57]. Their design must account for source impedance, waveform, voltage, and available power. An interface that performs well under steady sinusoidal excitation may behave differently during startup or isolated mechanical events.

For piezoelectric sources, the internal capacitance makes charge management particularly important. Synchronous switching can modify the voltage across that capacitance, changing the interaction between charge generation and extraction. The nonlinear processing introduced by Guyomar et al. [58] and the bias-flip rectifier developed by Ramadass and Chandrakasan [59] established influential approaches. These methods also show why the interface cannot always be treated as a fixed efficiency applied to independently measured transducer power: switching changes the electrical boundary and can alter mechanical motion.

Consequently, reported gains require a defined reference circuit, input, and load. Extraction improvement relative to a bridge rectifier does not establish an equivalent increase in energy reaching storage. Switching, conduction, timing, and control losses remain within the assessment boundary, particularly when the source provides only intermittent or weak excitation.

7.2 Integrated Interfaces and Co-Design

Recent interfaces increasingly coordinate circuit architecture with transducer structure. Yue et al. [60] combined split electrodes with a fully integrated sequential synchronised-switch harvesting-on-capacitors rectifier. Salem [61] investigated a recursive N-path switched-capacitor rectifier, addressing charge redistribution through another circuit architecture. Both illustrate alternatives to treating the transducer and rectifier as independently selected components. Their applicability nevertheless depends on the capacitance, voltage, and operating conditions of the connected source.

Component sharing provides a further integration route. Wei et al. [62] developed a synchronous piezoelectric–electromagnetic platform using a shared coil. Sharing can reduce duplicated components, but the integrated circuit, mechanical assembly, storage, and load retain distinct physical boundaries. A circuit-level integration result should therefore identify which external components remain necessary.

The evidence supporting each architecture also matters. Shoorabi Sani [63] investigated an impact-oriented interface through post-layout complementary metal–oxide–semiconductor (CMOS) circuit simulation; those results describe a circuit design rather than measured operation of fabricated silicon. Verma et al. [64] considered combined piezoelectric and triboelectric sources, for which different source characteristics require coordinated

power management. Iwanari et al. [65] investigated a self-powered bias-flip rectifier with on-chip timing control, bringing autonomous switching operation into the interface design.

Co-design can extend beyond the electrical circuit. Gu et al. [66] reused a magnetic core within a piezoelectric harvesting and extraction arrangement for transmission-line monitoring. Its input was an alternating magnetic field rather than base vibration, so it informs component-sharing principles without providing a directly comparable vibration-power result. Together, these studies support assessment of the complete component set and its operating requirements, rather than circuit area alone.

7.3 Storage and Leakage

Storage connects intermittent generation to sensing tasks whose timing and peak demand may differ from the input. The review of flexible micro-supercapacitors by Pantrangi et al. [67] describes material and architectural options, but storage selection also requires usable voltage range, leakage, resistance, and package volume. Electrode-level capacitance alone does not determine node operating time.

For an ideal capacitor, the energy available between two voltage limits is:

$$E_{\text{use}} = \frac{1}{2} C_{\text{s}} \left(V_{\text{H}}^2 - V_{\text{L}}^2 \right) \tag{8}$$

where, C_{s} is capacitance, V_{H} is the upper operating voltage, and V_{L} is the minimum permitted voltage. Eq. (8) assumes constant capacitance and excludes leakage, resistive losses, and downstream conversion losses. It describes an energy window, not guaranteed energy at the sensing load.

Yamada et al. [68] combined modelling and experiments for a vibration-powered wireless node with a built-in supercapacitor, providing a system-level basis for examining these interactions. Storage must support task completion as well as average demand. Increasing capacitance can bridge longer interruptions, but it also increases the energy required to reach a given startup voltage. Load control therefore belongs within storage design. The three-transistor management circuit investigated by Teng et al. [69] addresses this connection through controlled operation of an intermittently powered load.

7.4 Net Energy and Cold-Start Requirements

Cold start and sustained operation impose different requirements. A circuit may operate efficiently after its storage voltage rises sufficiently yet fail to accumulate the initial energy under the intended input. Conversely, reaching a startup threshold does not ensure completion of the next sensing or transmission task. Assessment should specify the initial storage state, excitation history, startup time, and behaviour during repeated load activation.

Figure 5 relates startup and repeated task execution to the measurements needed to establish net energy delivery. The storage-voltage trace illustrates how task demand and reduced input can interrupt operation even after successful startup. Loaded-source output, storage-energy balance, and completed sensing service must therefore be verified at their respective boundaries.

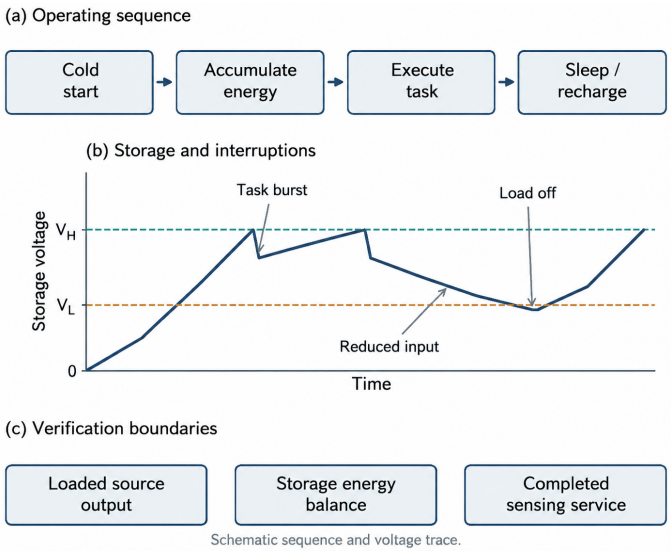


Figure 5. Conceptual startup sequence, storage-voltage evolution, and verification boundaries (curves are schematic)

These conditions complete the energy balance introduced in Eq. (2). Net-energy reporting should identify where power was measured, which control and conversion losses were included, and whether stored energy increased or decreased during node operation. Wireless activity supported partly by previously stored energy must be distinguished from operation sustained by the contemporaneous input.

Zhao et al. [70] integrated harvesting, power management, sensing, and wireless communication in an assembled system under controlled periodic excitation. Such evidence connects more of the energy path than a resistor-load measurement, while its excitation and assembly scale remain necessary qualifications. Table 5 summarises the evidence required at each stage. The next section examines whether application studies establish these conditions in their intended deployment environments.

Table 5. Interface, storage, and system evidence relevant to net energy delivery

Assessment Stage	Representative Studies	Information Needed for Interpretation
Rectification and extraction	[58–61]	Source and load conditions, reference circuit, electrical boundary, and control losses
Shared components and hybrid inputs	[62, 64, 66]	Input type, external components, sharing constraints, and complete assembly
Interface simulation	[63]	Simulation assumptions and distinction from fabricated-circuit measurements
Autonomous switching and startup	[65]	Initial energy state, startup conditions, timing consumption, and repeated operation
Storage integration	[67, 68]	Usable voltage window, leakage, resistance, charging history, and load demand
Intermittent load management	[69]	Activation thresholds, task completion, standby consumption, and sensing rate
Integrated wireless sensing	[70]	Excitation, test duration, storage-energy change, achieved service, and physical scale

8 Application Evidence and Deployment Readiness

8.1 Energy Equipment and Environmental Nodes

The energy path examined in Section 7 provides a basis for assessing applications: a useful demonstration must connect the available excitation to a specified sensing service. For energy equipment, this requires establishing whether vibration persists across the operating states in which measurements are needed. A node supplied by machine vibration may lose its input when the machine stops, even though monitoring remains necessary. Storage and task scheduling must therefore reflect the monitoring requirement as well as the normal operating condition.

The transformer temperature-monitoring system investigated by Wang et al. [12] directly addresses an energy-sector application. Its low-acceleration harvesting and event-triggered operation illustrate how a sensing task can be matched to a restricted supply. The demonstrated service must nevertheless be described through its measurement and transmission behaviour; event-triggered operation does not imply continuous acquisition or unrestricted reporting.

Environmental-node studies provide complementary evidence about the supporting energy system. Bouhedma et al. [7] examined battery-assisted and battery-free operation under controlled harmonic excitation. Their results help distinguish reduced battery demand from operation without a battery, but projected battery-life extensions remain different from measured service life. Zhao et al. [70] connected harvesting, power management, and wireless sensing within an assembled system. This establishes a more complete functional chain, while leaving the effects of variable field excitation and installation conditions to be evaluated separately.

Deployment also introduces communication demands that an isolated node test may not capture. Van Leemput et al. [71] examined the integration of batteryless devices into multihop industrial wireless networks. Their analysis brings energy availability into the same design problem as communication timing and reliability. A node may complete a local measurement yet lack sufficient energy to participate in the required network schedule. Application assessment should therefore report successfully delivered information, rather than sensing events alone.

8.2 Transport Infrastructure

Transport infrastructure provides a related setting in which excitation is spatially variable and associated with discrete traffic events. Hybrid vibration harvesting for railway sensing was investigated by Zhu et al. [72], extending the application discussion beyond individual transducers. As a larger system, it informs the coupling of harvesting and sensing functions without establishing the performance of a MEMS equivalent.

Field measurements are particularly valuable in this setting because installation position influences the available input. Ordoñez et al. [73] combined experimental and numerical assessment of an electromagnetic harvester in a double-deck railway tunnel. The measured mechanical and electrical responses provided site-specific evidence, whereas cumulative-energy estimates relied on a calibrated model. These outputs answer different questions: the measurements characterise the tested response, while the model extends the assessment across the specified input record. Neither should be presented as a direct record of long-term wireless-node operation.

Sheng et al. [74] investigated a piezoelectric-powered wireless monitoring system for railway bridges, bringing the assessment closer to the sensing service. This complements the bridge study discussed in Section 3 [13], which used measured vibration as input to numerical harvesting and sensing optimisation. Taken together, the studies show why realistic input data, a functioning harvester, and a powered monitoring system should be identified separately. Larger infrastructure devices can establish relevant design principles, but their mass, travel, mounting, and packaging requirements constrain transfer to MEMS.

8.3 Deployment Readiness and Evidence Limits

The reviewed applications support different conclusions according to what was tested. A model driven by field data can examine sensitivity to a realistic input. Field measurements of a harvester can establish its response at a particular installation. A complete-node demonstration can additionally establish task execution and communication. Sustained deployment requires evidence that these functions remain available as excitation, storage state, and environmental exposure change.

Table 6 retains these distinctions. Comparisons should identify the physical scale, excitation source, test duration, initial storage condition, and achieved sensing service. Where the verification did not establish a deployment duration or complete energy balance, this review does not infer one from device output or the description "self-powered."

Table 6. Application evidence and the conclusions it supports

Application and Reference	Evidence Considered	Supported Interpretation and Remaining Boundary
Transformer temperature monitoring [12]	Low-acceleration harvesting with wireless monitoring	Application-specific feasibility; event-triggered service remains distinct from continuous acquisition
Environmental sensing [7]	Controlled harmonic tests of battery-assisted and battery-free operation	Operating scenarios and measured energy demand; projected lifetime is not observed field life
Integrated wireless sensing [70]	Assembled harvesting, management, and sensing system	Functional-chain demonstration under controlled periodic input
Industrial multihop networks [71]	Batteryless-node integration strategies	Communication and scheduling requirements; not mechanical-harvester endurance evidence
Railway sensing [72]	Hybrid harvesting and sensing architecture	Cross-scale application reference; not direct MEMS performance evidence
Railway tunnel [73]	Field responses combined with calibrated modelling	Site-specific characterisation; measured responses and modelled cumulative energy remain separate
Railway bridges [13, 74]	Field-data-driven optimisation and wireless-system investigation	Complementary modelling and application evidence; no general inference of maintenance-free service

Note: MEMS = micro-electro-mechanical system.

Readiness consequently depends on maintaining the required service within a documented operating envelope. That envelope should include inactive periods, restart behaviour, output degradation, and the consequences of reducing the sensing rate. These conditions also determine whether harvesting can reduce battery replacement or maintenance. Section 9 uses them to examine the life-cycle assumptions behind claims of environmental improvement.

9 Life-Cycle Sustainability and Research Priorities

9.1 Functional Units and System Boundaries

The application evidence in Section 8 establishes the operating conditions that a sustainability assessment must retain. A harvester has environmental value when it provides a specified sensing service with lower life-cycle burdens

than a credible alternative. Device output, reduced battery capacity, and maintenance avoidance contribute to that comparison, but none establishes the outcome independently.

The functional unit should therefore describe the monitoring service, including its duration, measurement frequency, communication requirements, and acceptable data loss. Comparing devices per unit mass can inform material selection, while comparing impacts per unit of generated electricity can inform conversion assessment. Neither necessarily represents the service delivered by a sensor node. A harvesting system that achieves a lower environmental burden by reducing measurements cannot be compared directly with a battery-powered node providing a more demanding service.

Material substitution illustrates the importance of this distinction. Royo-Silvestre et al. [75] investigated ferrite substitution in an ultra-low-frequency magnetic-levitation harvester. Their environmental discussion drew on existing magnet assessments and the quantities used in the prototypes; the moving magnet remained NdFeB. The study consequently informs a specific substitution rather than establishing a rare-earth-free harvester or a complete node-level life-cycle assessment. Amato et al. [76] provided upstream evidence through production-stage assessment of rare-earth-free magnet alternatives. Their mass-based functional unit does not establish equivalence in magnetic performance, device dimensions, or delivered energy.

A node-level boundary must extend beyond the active material to fabrication, electrodes, proof masses or magnets, packaging, interface electronics, storage, transport, replacement, and end-of-life treatment. Where components are identical between alternatives, their treatment should remain explicit. Zhang et al. [77] examined circularity through integrated-circuit recycling and alternative electronic assembly materials, demonstrating the relevance of electronic components and recovery assumptions. Their consumer-electronics cases inform the treatment of electronic components and recovery assumptions in MEMS-node assessments.

9.2 Environmental and Maintenance Break-Even

For alternatives providing the same service over an assessment period T, the difference in environmental impact can be expressed as:

$$\Delta I_j(T) = I_{\mathrm{H},j}(T) - I_{\mathrm{R},j}(T) \tag{9}$$

where, $I_{\mathrm{H},j}$ is the life-cycle impact of the harvesting configuration, $I_{\mathrm{R},j}$ is that of the reference configuration, and j identifies an impact category. Each total includes production, use, replacement, maintenance, and end-of-life contributions within the declared boundary. A negative value indicates an advantage for harvesting in that category under the stated assumptions. Eq. (9) is an accounting relation; it does not imply that impacts accumulate linearly or that a single break-even time applies across categories.

The reference configuration requires particular care. A battery-only node, a battery-assisted harvesting node, and a battery-free node have different component inventories and replacement schedules. The comparison must use realistic battery capacity, standby consumption, reporting behaviour, and service life. Environmental credit for avoided replacement is justified only when harvesting prevents an otherwise necessary intervention. Several nodes may be serviced during one visit, and battery replacement may coincide with other maintenance. Node count therefore cannot be converted directly into avoided journeys.

Harvester durability is equally consequential. Film degradation, loss of electret charge, packaging failure, or storage ageing can change replacement requirements before the intended service period ends. A short endurance test supports behaviour under its tested conditions; extending it to a deployment lifetime requires an explicit model and uncertainty analysis. The material assessments discussed earlier [8, 9] likewise retain their original boundaries and lifetime assumptions rather than supplying a universal environmental payback period.

Production location and disposal practice introduce further variation. Sharma et al. [78] assessed recycled plastic used in triboelectric nanogenerator manufacture under Indian and Singaporean scenarios. Their gate-to-grave boundary and regional assumptions constrain interpretation of the results. Zikulnig et al. [79] examined environmental burdens in printed sensors, while excluding the external radio-frequency reader power supply. This exclusion matters when moving from a sensor assessment to a complete monitoring system. Recycling, passive operation, or material substitution should thus be evaluated within the functions and processes actually included.

9.3 Evidence Gaps and Research Priorities

The reviewed literature supports a progression from material and device evidence towards service-based assessment. The first priority is to connect representative vibration histories with measured electrical output and achieved sensing workloads. Reporting only a favourable resonant operating point leaves the use-phase contribution unresolved. Tests should identify storage-energy changes, interruptions, restart behaviour, and completed communication tasks so that functional equivalence can be assessed.

A second priority is to connect performance degradation with replacement assumptions. Reliability studies should report the loading history, exposure, failure criterion, and changes in output, while lifecycle models should

show how those observations affect service availability and component replacement. This connection is needed before laboratory endurance can support claims about reduced maintenance.

A third priority concerns manufacturing inventories. Material quantities, deposition and release processes, device yield, packaging, and assembly should be documented at boundaries that allow meaningful comparison. Changes described as lead-free, water-assisted, printed, or recyclable require assessment of the remaining processes and components. Their value may differ across environmental categories, making sensitivity analysis more informative than a single unqualified "green" designation.

Doyle et al. [80] applied green-engineering principles to a low-technology-readiness energy-harvesting platform, providing a framework for incorporating such questions during development. Early design assessment can guide choices before production data are complete, provided measured inventories, estimates, and prospective scenarios remain distinguishable. Table 7 summarises the evidence required. The resulting research objective is to establish when a particular harvesting system sustains the required service with fewer life-cycle burdens, and how securely that conclusion follows from the available data.

Table 7. Sustainability evidence and limits on transfer to MEMS sensor systems

Evidence Source	Assessed Object or Approach	Requirement for Node-Level Interpretation
Piezoelectric assessments [8, 9]	PZT components, road systems, and active-film alternatives	Retain functional units, lifetime assumptions, and excluded components
Magnet substitution [75]	Prototype-specific material substitution using existing environmental evidence	Include the remaining magnets and consequences for device function
Magnet production [76]	Mass-based assessment of alternative magnet manufacture	Establish functional equivalence before comparing harvesters
Circular electronics [77]	Integrated circuits and assembly materials	Adapt inventories and recovery assumptions to the actual node
Recycled-plastic TENG manufacture [78]	Regional gate-to-grave scenarios	Retain geographic and process boundaries
Printed sensors [79]	Sensor materials and processing	Include external equipment where required by the monitoring service
Green engineering [80]	Design assessment at low technological maturity	Distinguish design guidance from measured product-level outcomes

Note: MEMS = micro-electro-mechanical system; TENG = triboelectric nanogenerator; PZT = lead zirconate titanate.

10 Conclusions

MEMS vibration energy harvesting can support sustainable sensor power when the available motion, transducer response, and sensing workload are matched. The reviewed evidence does not support a universal preference among the four conversion mechanisms. Their suitability depends on excitation, displacement limits, fabrication, and electrical loading. Device-level power is therefore an intermediate result; the relevant outcome is energy delivered to a specified sensing service after conditioning, control, and storage losses.

Reliable operation requires that this energy balance remain viable as inputs and device properties change. Nonlinear structures, tuning, and frequency up-conversion offer ways to accommodate frequency mismatch, but introduce operating thresholds, additional components, or repeated loading. Laboratory demonstrations establish feasibility within tested conditions, while deployment requires evidence covering interruptions, restart, degradation, and successful communication. Neither a self-generated sensing signal nor short-term wireless operation establishes long-term autonomy.

Environmental improvement requires a further comparison against an alternative providing the same service. Battery reduction can lower some burdens, but fabrication, packaging, electronics, storage, and replacement remain part of the assessment. Material substitutions and manufacturing changes must consequently be evaluated within explicit life-cycle boundaries rather than treated as environmental benefits in themselves.

The priority is to connect these three evidence levels: measured energy delivery, service continuity, and life-cycle performance. Representative input histories, complete energy accounting, degradation measurements, and transparent inventories would make that connection testable. Progress towards sustainable sensor power depends on

demonstrating this relationship across the intended operating life, with conclusions proportionate to the evidence.

Data Availability

Not Applicable. No new experimental data were generated in this study. The data supporting the findings of this study (literature-derived values plotted in Figure 3 and their reported test conditions) are included within the article (Table 2) and its cited references [23, 25, 32].

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the author used ChatGPT to improve grammar, wording, and readability. The author reviewed and revised the edited text and takes full responsibility for the final manuscript.

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Nomenclature

A	Base-acceleration amplitude for sinusoidal excitation
C	Capacitance of the electrostatic converter
$C(x)$	Configuration-dependent capacitance of the triboelectric converter
C_{p}	Effective piezoelectric capacitance
C_{s}	Storage capacitance
c_{m}	Mechanical damping coefficient
e	Induced electromotive force
E_{s}	Energy stored in the storage element
E_{use}	Ideal capacitive energy available between the operating voltage limits
f	Excitation frequency
F_{tr}	Signed transduction reaction term in the mechanical equation
g	Standard gravitational acceleration, 9.80665 m^{-2}
$I_{\mathrm{H},j}$	Life-cycle impact of the harvesting configuration in impact category k
$I_{\mathrm{R},j}$	Life-cycle impact of the reference configuration in impact category k
k	Effective modal stiffness in Eqs. (3) and (4)
j	Environmental impact-category index in Eq. (9)
m	Effective modal mass
P_{ch}	Power entering storage after upstream conversion losses
P_{draw}	Power withdrawn from storage for node operation, including downstream conversion and control consumption
P_{leak}	Power lost through storage leakage
Q	Capacitor charge in Eq. (5); transferred charge under the selected sign convention in Eq. (7)
R	External resistive load
t	Time
T	Life-cycle assessment period
U_{e}	Electrostatic energy stored in the converter capacitance
v	Piezoelectric terminal voltage in Eq. (4)
V	Capacitor voltage in Eq. (5); triboelectric terminal voltage in Eq. (7)
V_{H}	Upper operating voltage of the storage capacitor
V_{L}	Minimum permitted operating voltage of the storage capacitor
V_{oc}	Configuration-dependent open-circuit voltage
x	Displacement coordinate describing the triboelectric configuration
y	Base displacement
Y	Base-displacement amplitude for sinusoidal excitation
z	Relative displacement of the effective modal mass with respect to the base

Greek symbols

Γ	Magnetic flux-linkage gradient, dλ/dz
ΔI_j	Difference in life-cycle impact between harvesting and reference configurations in category j
θ	Modal piezoelectric coupling coefficient in Eq. (4)
λ	Magnetic flux linkage

Mathematical notation

$\dot{()}$	First derivative with respect to time
$\ddot{()}$	Second derivative with respect to time
d/dt	Differentiation with respect to time